Dwight Look College of
ENGINEERING
TEXAS A&M UNIVERSITY

ECEN 404 Final Presentation
Team 46: EchoPoint

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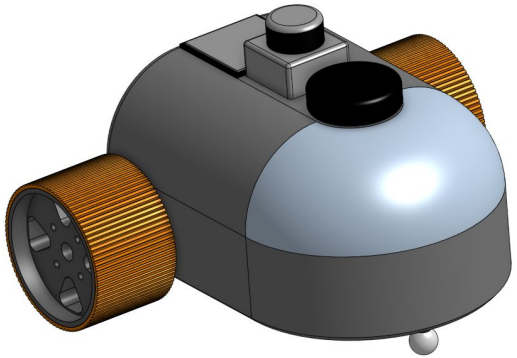
Project Overview

Problem Statement

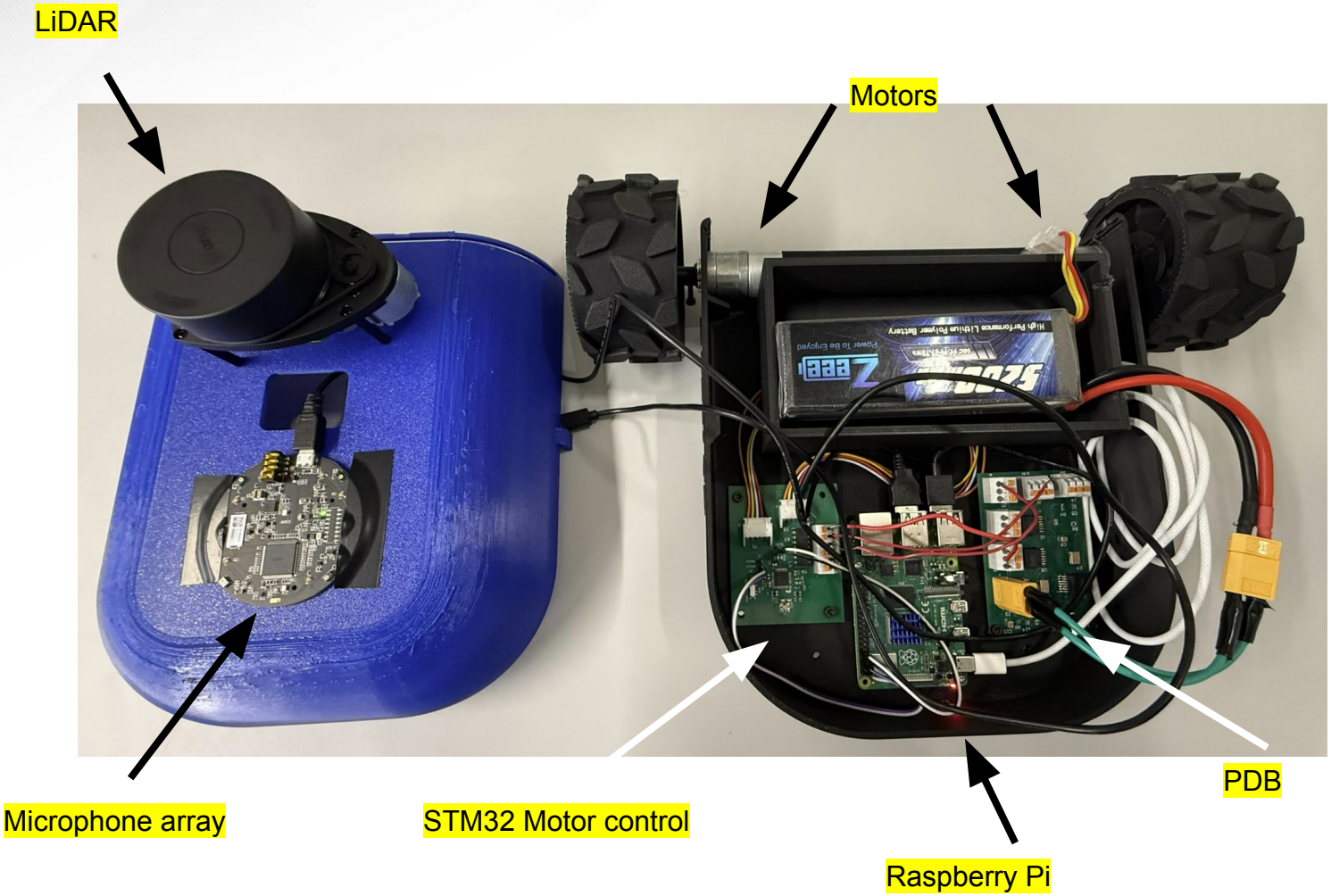
- Victims in rescue situations may be heard but not seen
- Camera-based robots fail when visibility is poor
- Need a system that locates people using sound instead of sight

Our Solution

- Mobile robot with microphone array for sound localization
- Raspberry Pi for real-time audio processing and direction detection
- STM32 microcontroller for motor control and navigation
- LiDAR for obstacle detection and safe path planning
- Robot moves toward clapping/voice source to help first responders locate individuals



Integrated Project Diagram

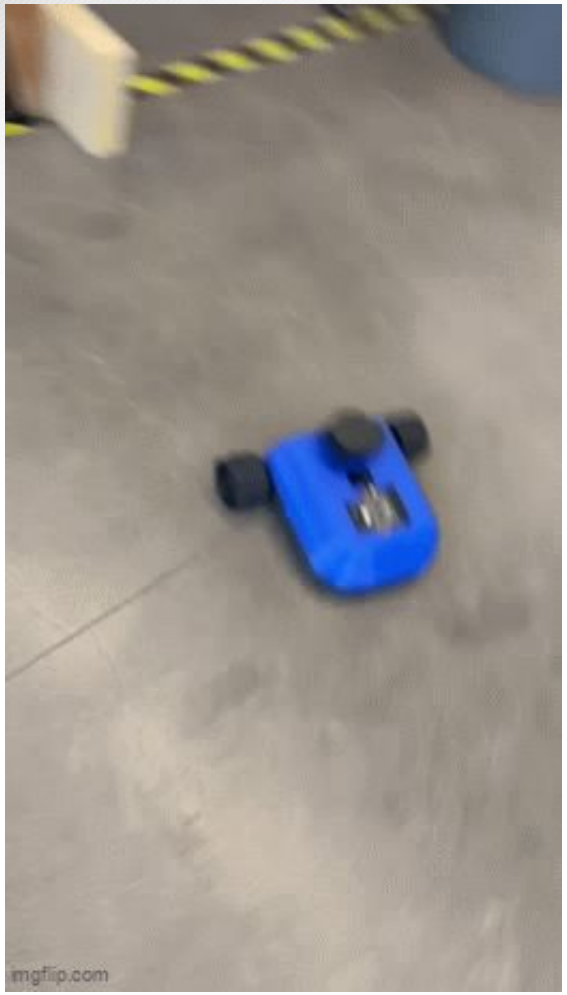


Motor Controller PCB and Chassis Accomplishments

Challenges on subsystems	Accomplishments on Echo Point
Chassis - Spacing for wires - Fell off table and cracked the chassis Motor PCB - MCU not turning on properly - MCU died and had to be replaced - Wires connecting to motors are weak - Reverse PWM on Motors - Not enough traction on wheels	- Chassis completely printed and holds everying needed in the robot - Motor PCB correctly interprets data from Pi and moves motors - Motor PCB is stable and has fail safes in case of connection loss to Pi Fail safe for motor PCB - When motors dont receive any signal for 300 ms they shut off

Motor Controller PCB and Chassis Accomplishments

Andrew Kazour



Move speed

- 3 ft/s
- Any smooth/slightly bumpy terrain
- PWM control for different speeds depending on situation

Motor reaction delay

- 15-50 ms

LiDAR Subsystem Accomplishments

Challenges on subsystems	Accomplishments on Echo Point
LiDAR Coding - Integration with Mic and Motor Code - LiDAR only had distance sensing - Realized obstacle avoidance logic had to be implemented in LiDAR code - Challenges implementing LiDAR override Speaker PCB - Attempted to replace original speaker with a different audio output - Speaker damage required rebuilding the PCB - Rebuilt speaker PCB now functions properly	- Successfully integrated microphone, LiDAR, and motor control code - Implemented functional distance sensing and obstacle avoidance - Achieved accurate distance measurements with completed validation - LiDAR system meets FSR requirements - Resolved prior issues preventing correct LiDAR operation - System now operates reliably with only minor tuning potentially required

LiDAR Subsystem

True Distance	LiDAR Sensing	Percent Error
0.25 m	0.27 m	≤1.08%
0.75 m	0.80 m	≤1.06%
1 m	1.03 m	≤1.03%
1.25 m	1.23 m	≤1.02%
4 m	3.89 m	≤1.02%
5 m	5.32 m	≤1.06%
6 m	6.02 m	≤1%
7 m	6.84 m	≤1.02%
8 m	8.03 m	≤1%

Percent Error after 60 trials

Power Subsystem Accomplishments

Arabella Aguirre

Challenges on subsystems	Accomplishments on Echo Point
- PCB did not initially output the correct voltage required for the STM32. - During testing, PCB experienced a short circuit that prevented proper operation - The updated PCB design was intended to provide stable output and better current delivery to the STM32	- Successfully fixed PCB by replacing faulty IC chips, restoring proper functionality - Verified that the power voltage was distributed and ensured reliable electrical operation. - Resolved previous voltage regulation issues that were preventing the PCB from working correctly

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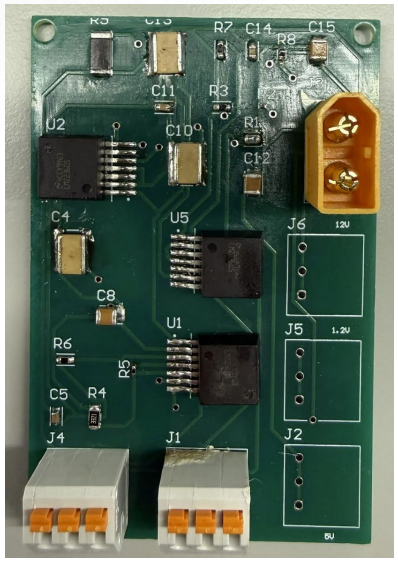
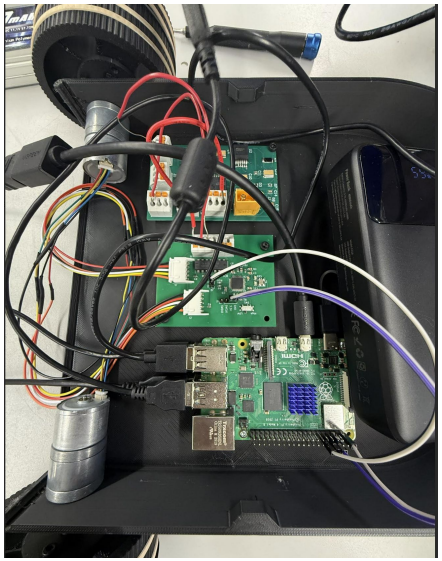
Power Subsystem

Arabella Aguirre

Echo Point Functiong = Power successful



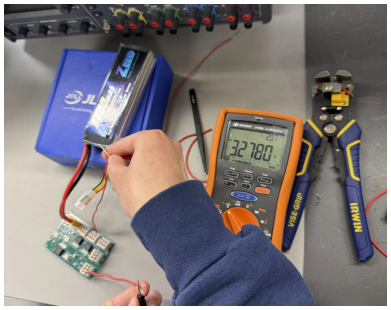
PCB = fully soldered



PCB = fully integrated with robot

12.0V

Raspberry Pi = 5.0

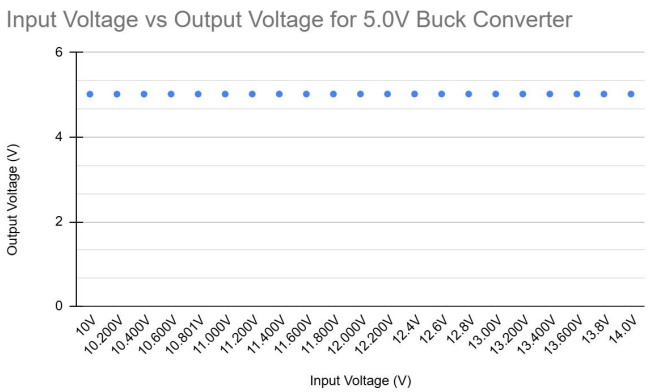


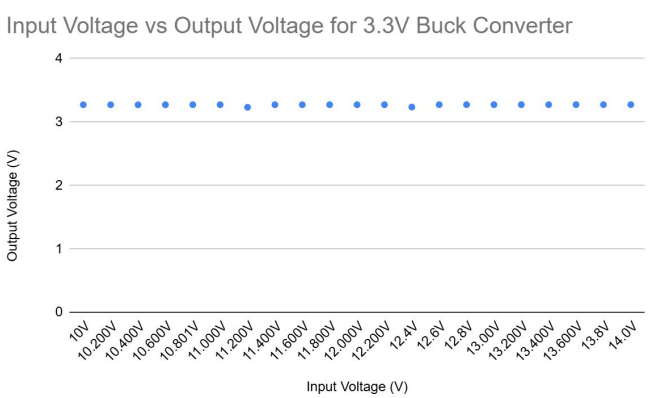
STM32 = 3.2V

Power Subsystem

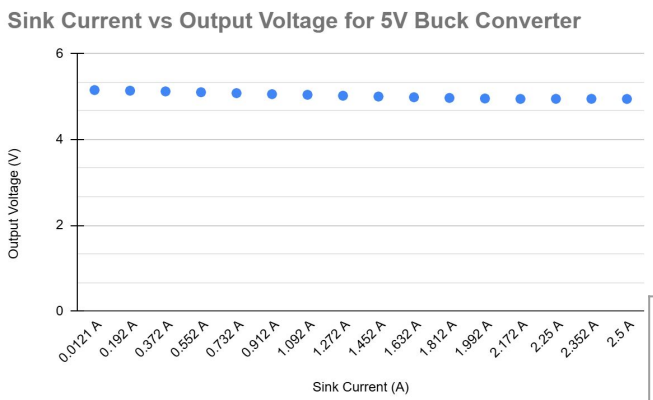
Arabella Aguirre

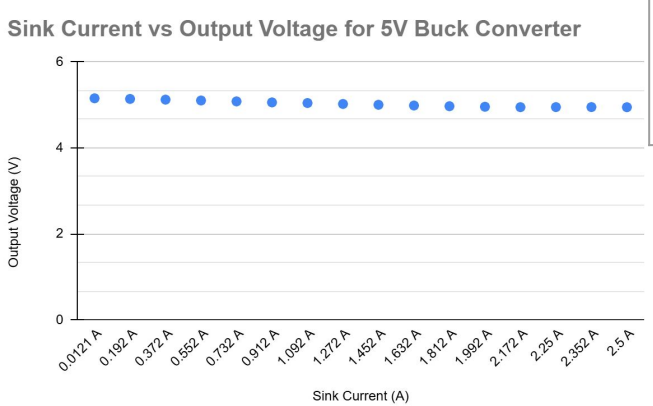
Open Circuit/LineTesting





Electronic Load Test





Battery life of motors	Raspberry Pi and peripherals
3 hours	3 hours

Microphone Subsystem Accomplishments

Challenges on subsystems	Accomplishments on Echo Point
- Getting the microphone subsystem to calculate the correct angle - Finding a sound that the microphone subsystem could both detect and calculate the correct direction - Rooms with lots of echo, noisy rooms, sound of motor distracted the robot - When integrated, robot found it hard deciding whether to prioritize LiDAR or microphone code	- Altered microphone code to accurately calculate angle from where sound is coming from, given real time audio data - Successfully integrated microphone subsystem with LiDAR, motor controller, and power subsystems

Microphone Subsystem

Distance	Sound Detected?	Average Angle Error (degrees)
0.25 m - front (0 degrees)	Yes	±6
0.75 m - front (0 degrees)	Yes	±7
1 m - front (0 degrees)	Yes	±10
3 m - front (0 degrees)	Yes	±11
0.25 m - side (90 degrees)	Yes	±13
0.75 m - side (90 degrees)	Yes	±18
1 m - side (90 degrees)	Yes	±8
3 m - side (90 degrees)	Yes	±16

- Furthest distance robot can detect sound: 10 ft

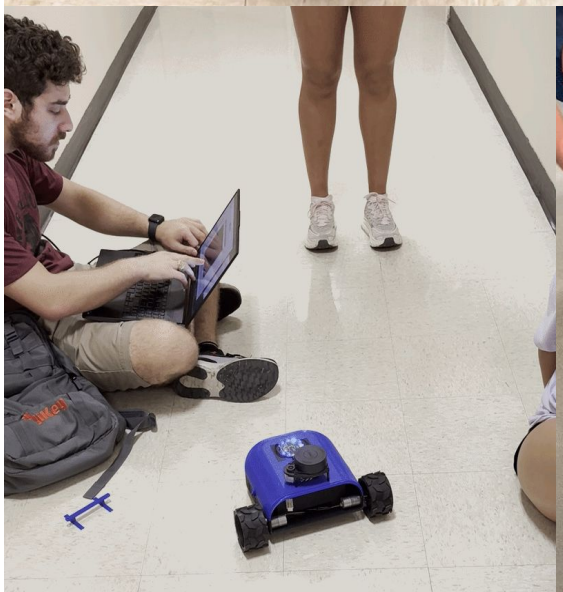
Microphone Subsystem

Distance	Sound Detected?	Average Angle Error (degrees)
0.25 m - back (180 degrees)	Yes	±20
0.75 m - back (180 degrees)	Yes	±20
1 m - back (180 degrees)	Yes	±25
3 m - back (180 degrees)	Yes	±8

Integrated System Results

Follow mode

- Follows with 80% accuracy
    - Out of 50 trials it messed up on10 trials
- Follows at 3 ft/s
- Avoids obstacles with a 84% accuracy
    - Out of 50 trials it hit 8 time with the wheel of the robot



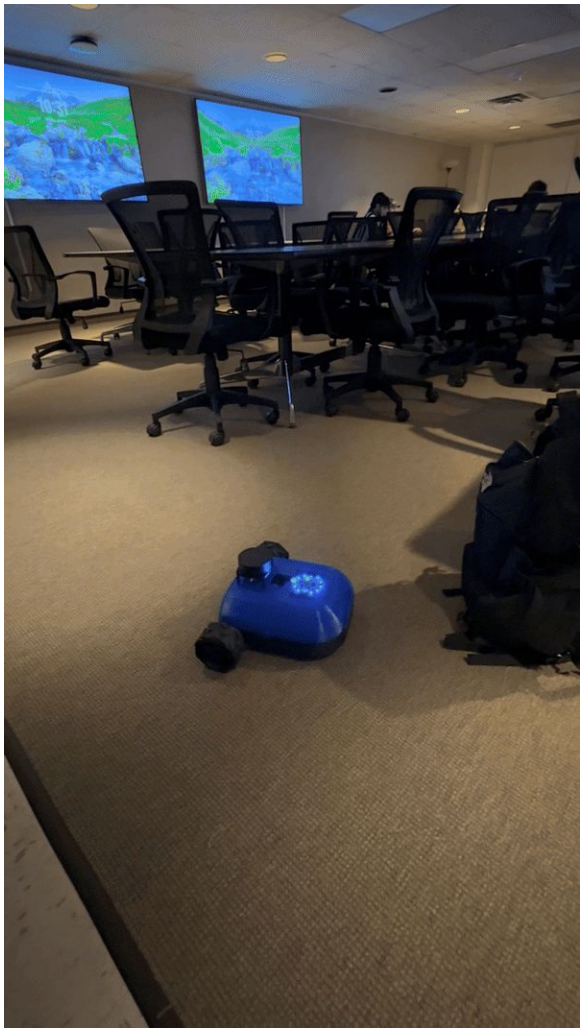
Integrated System Results

Longer range tracking mode

- 10 ft accurate tracking range

Environment

- Echoey environments
  - Takes on average 2x as long to find the sound source
- Normal/quiet environment
  - Lands within 2 feet of sound source



System Validation Plan (Pt. 1)

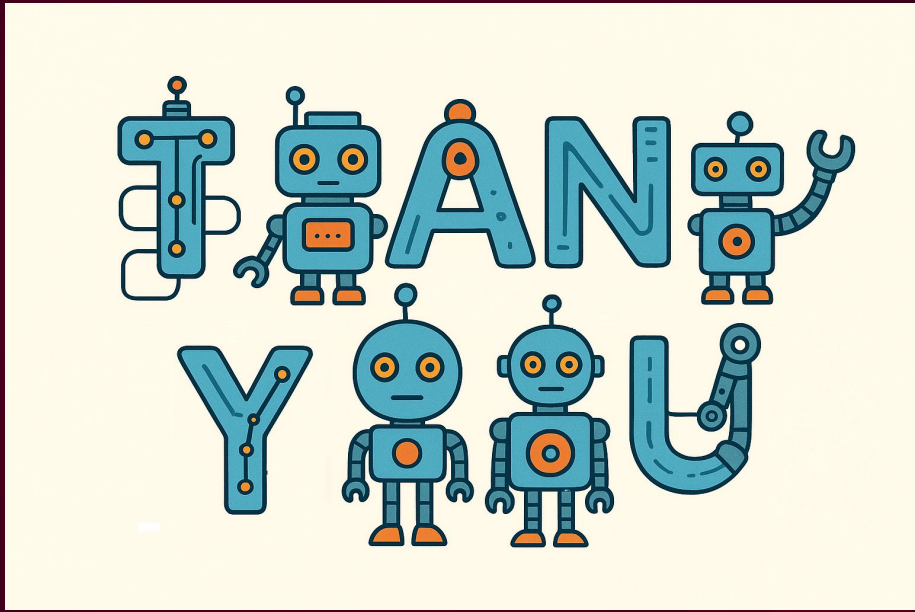
Test Name	Success Criteria	Methodology	Status	Responsible Engineers
3.2.8.2. Diagnostic Output	The Search and Rescue System shall include a diagnostic interface remotely for control and data logging.	The diagnostic interface will be tested by seeing if proper mic, heading angle, and Lidar scans are correct. Issuing manual control commands and downloading system logs confirm proper debugging control and data logging functionality.	Complete	All
3.2.10.4. BUS False Alarms	The Robot shall have a false alarm rate of less than 10 percent.	The system will operate under normal conditions for extended trials and verifying the false alarm rate remains below 10%.	Complete	All
3.2.10.5.Isolation and Recovery	The Search and Rescue System should provide for recovery by enabling subsystems to be reset or disabled based upon the result of the BUS.	Subsystem faults will be triggered and the Robot response will be observed to verify that affected subsystems can be reset or disabled without interrupting overall system operation.	Complete	All

System Validation Plan (Pt. 2)

Test Name	Success Criteria	Methodology	Status	Responsible Engineers
3.2.9.3. Terrain	The Search and Rescue System will be able to operate on smooth ground or an incline.	The robot will be tested on smooth surfaces and inclined ramps while executing search operations to verify stable mobility and proper system functionality.	Complete	All
3.2.10.2. Back up system (BUS)	The Search and Rescue System shall check its sound localization with the LiDAR data to ensure it is going in the correct direction. Cross referencing this data will lead to less error propagation.	Sound localization results will be cross-checked with LiDAR environmental mapping during operation to confirm the system validates navigation direction using redundant sensor data.	Complete	Amber, Andrew, Melissa
3.2.10.3. Robot Sound Precision	The Robot should be able to detect a sound with 90% accuracy and go towards it in real time.	Play sounds aroud the robot and test how many times it properly goes towards the sound.	Complete	Amber, Andrew, Melissa

Conclusion

- Successfully developed an integrated sound-localization search robot using microphone, LiDAR, motor, and power subsystems
- System required multiple design adjustments during development, including LiDAR control logic updates, PCB redesigns, and subsystem integration fixes
- LiDAR obstacle avoidance and microphone sound-tracking were successfully integrated to enable safe autonomous navigation
- Testing verified accurate distance sensing, sound detection, and obstacle avoidance performance
- System validation tests for power, sensing, terrain mobility, and sensor cross-checking were successfully completed
- Final system operates reliably and demonstrates the feasibility of sound-based search assistance for rescue scenarios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